

Problem Statement

Modern System-on-Chip (SoC) designs consume a significant amount of leakage power even when functional blocks remain idle. Conventional power-gating techniques typically rely on static ON/OFF policies, which either keep a block fully powered or completely shut it down. Although effective in reducing leakage, these approaches often introduce unnecessary wake-up overhead and fail to efficiently handle varying workload patterns. Therefore, there is a need for a lightweight adaptive power-management mechanism capable of reducing leakage while maintaining acceptable performance and hardware overhead.

Motivation

As technology scales into deep-submicron nodes, leakage power becomes an increasingly dominant component of total chip power. Many embedded processors and digital accelerators spend a considerable amount of time in idle or partially active states, making static power management inefficient. A lightweight adaptive controller that dynamically selects different sleep modes can improve the trade-off between leakage power and wake-up latency without requiring complex prediction algorithms. This motivates the development of lightweight hardware-based adaptive techniques that can intelligently manage power states with minimal implementation overhead.

Research Objective

The objective of this work is to design and evaluate a lightweight adaptive multi-level power-gating controller for a small digital block. The controller monitors hardware activity and dynamically transitions between ON, LIGHT SLEEP, and DEEP SLEEP states using a lightweight activity-history-based hysteresis decision mechanism. The proposed design is evaluated through RTL implementation, functional verification, synthesis, place-and-route, and power analysis using the Synopsys design flow.

Scope

This work focuses on RTL-level design and evaluation of the power-management controller. The functional block selected for evaluation is a 32-bit ALU. Software-based power management, operating-system scheduling, and dynamic voltage/frequency scaling (DVFS) are outside the scope of this project.

Expected Contributions

1. A lightweight adaptive multi-level power-gating controller for small RTL-based digital systems.
2. A lightweight activity-monitoring mechanism for estimating block utilization.
3. A hysteresis-based decision policy that reduces unnecessary sleep-state oscillations.
4. A complete RTL implementation validated using the Synopsys RTL-to-GDSII design flow.

5. A quantitative comparison against a conventional ON/OFF power-gating controller using identical implementation constraints.

Assumptions

Same technology library for all experiments.

Same synthesis constraints.

Same operating voltage.

Same clock frequency unless otherwise stated.

Same ALU benchmark.